

Random Walk and the Boundary Value Problems

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Theorem (Black–Scholes Equation).

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0.$$

- The **random walk model** serves as the foundation for the Black-Scholes equation.
- My work is primarily based on Sections 1.1, 1.2, and 1.4 of G. Lawler's book, *Random Walk and the Heat Equation* [1]. In solving boundary value problems with the random walk models, I also referred to the article *Random Walks in $\mathbb{Z}^d$ and the Dirichlet Problem* by E. Guan [2].

Agenda

- 1 (Karl Pearson, 1905) **Construction**: a walker taking n successive equal steps in a two-dimensional plane, with an equal probability of moving in all directions [3].
- 2 (Lord Rayleigh, 1905 and later) The **probability distribution** for large enough values of n [4]. The solution for the three-dimensional case [5].
- 3 (George Pólya, 1921) A simple random walk exhibits **recurrent** behaviour for one- and two-dimensional cases, while **transient** behaviour for higher-dimensional cases [6]. (Shizuo Kakutani) “A drunk man will find his way home, but a drunk bird may get lost forever [7].”
- 4 Random walks found relevance in areas such as Brownian motion, and the heat equation, profoundly impacting various branches of mathematics and physical sciences [8]. A remarkable mathematical application of the random walk model is to solve the **Dirichlet problem**.

One-dimensional Random Walks

The most fundamental random walk models: the **simple random walk on the integer lattice $\mathbb{Z}^d$** (H. N. V. Temperley, [9]). A lattice is a array of points in an n -dimensional space, resembling a vector space where the vectors in a lattice can only be multiplied by integers [10].

Definition.

Denote the **position** of the random walker at time n by S_n , which can express S_n as the sum of the individual steps:

$$S_n = X_1 + \dots + X_n,^a$$

where $X_i = \pm 1$ are **independent and identically distributed** (i.i.d.) random variables denoting the change in position between time $i - 1$ and time i , with each step taken having equal probability

$$\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}.$$

^aWe take the starting point to be the origin for convenience; however, it can be chosen arbitrarily.

Properties

■ $\mathbb{E}[S_n] = 0.$

■

$$\mathbb{E}[S_n^2] = \mathbb{E}\left[\left(\sum_{j=1}^n X_j\right)^2\right] = \mathbb{E}\left[\sum_{j=1}^n \sum_{k=1}^n X_j X_k\right] = n + \sum_{j \neq k} \mathbb{E}[X_j X_k].$$

We have $\sum_{j \neq k} \mathbb{E}[X_j X_k] = 0$, as the probabilities of $X_j X_k = 1$ and $X_j X_k = -1$ are equal among the four random combinations of X_j and X_k being ± 1 . Thus, $\text{Var}(S_n) = n$, i.e., the **typical distance** of the random walker from the origin is of order $\sqrt{n}$. After n steps, the random walker is expected to be approximately $\sqrt{n}$ steps away from its starting point.

Central Limit Theorem

- Consider a random walker who takes $2n$ steps and returns to the origin. Then the random walker must take n steps both to the left and to the right, as each move takes one unit of length. Consequently, the remaining n steps must be leftward, and the probability of this occurrence is 2^{-2n} . Thus, we have

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 2^{-2n} = \frac{(2n)!}{n!n!} 2^{-2n}.$$

- More generally, consider the probability of our random walker reaching a position of $2j$. Choose $n + j$ steps to the right and $n - j$ steps to the left:

$$\mathbb{P}(S_{2n} = 2j) = \binom{2n}{n+j} 2^{-2n} = \frac{(2n)!}{(n+j)!(n-j)!} 2^{-2n}. \quad (1)$$

Stirling's Formula

Theorem (Stirling's Formula).

As $n \rightarrow \infty$,

$$n! \sim \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n},$$

where $\sim$ indicates that the ratio of the two sides converges to 1.

Proof. Use Wallis' formula. □

Applying Stirling's Formula to equation (1) gives

$$\begin{aligned} \mathbb{P}(S_{2n} = 2j) &\sim \frac{\sqrt{2\pi}(2n)^{2n+\frac{1}{2}} e^{-2n}}{\sqrt{2\pi}(n+j)^{n+j+\frac{1}{2}} e^{-(n+j)} \sqrt{2\pi}(n-j)^{n-j+\frac{1}{2}} e^{-(n-j)}} 2^{-2n} \\ &\sim \frac{1}{\sqrt{\pi}} \left(1 - \frac{j^2}{n^2}\right)^{-n} \left(1 + \frac{j}{n}\right)^{-j} \left(1 - \frac{j}{n}\right)^j \left(\frac{n}{n^2 - j^2}\right)^{\frac{1}{2}}. \end{aligned} \quad (2)$$

- When $j = 0$,

$$\mathbb{P}(S_{2n} = 0) \sim \frac{1}{\sqrt{\pi}} n^{-\frac{1}{2}}.$$

This result suggests that $\mathbb{P}(S_{2n} = 2j)$ also has a form of order $\sqrt{n}$.

- Substitute $j = r\sqrt{n}$ into the right-hand side of equation (2),

$$\frac{1}{\sqrt{\pi n}} \left(1 - \frac{r^2}{n}\right)^{-n} \left[\left(1 + \frac{r}{\sqrt{n}}\right)^{\sqrt{n}}\right]^{-r} \left[\left(1 - \frac{r}{\sqrt{n}}\right)^{-\sqrt{n}}\right]^{-r} \left(\frac{1}{1 - \frac{r^2}{n}}\right)^{\frac{1}{2}},$$

which is asymptotic to

$$\frac{1}{\sqrt{\pi n}} e^{r^2} e^{-r^2} e^{-r^2} = \frac{1}{\sqrt{\pi n}} e^{-\frac{r^2}{n}}.$$

as $n \rightarrow \infty$.

- For every $a < b$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(a\sqrt{2n} \leq S_{2n} \leq b\sqrt{2n} \right) = \lim_{n \rightarrow \infty} \sum \frac{1}{\sqrt{\pi n}} e^{-\frac{j^2}{n}}, \quad (3)$$

where the sum is taken over all j such that $a\sqrt{2n} \leq 2j \leq b\sqrt{2n}$.

- The right-hand side of this expression represents a Riemann sum approximation of an integral, with intervals of length $\sqrt{\frac{1}{\pi n}}$. Then this limit converges to the integral

$$\int_a^b \frac{1}{C} e^{-\frac{x^2}{2}} dx,$$

where C is a constant.

- To ensure that the limiting distribution is indeed a probability distribution, we must have

$$\int_{-\infty}^{\infty} \frac{1}{C} e^{-\frac{x^2}{2}} dx = 1.$$

By evaluating the integral, we find that $\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$.
Then

$$\int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx.$$

Central Limit Theorem.

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(a\sqrt{2n} \leq S_{2n} \leq b\sqrt{2n} \right) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx.$$

Central Limit Theorem for One-dimensional Random Walks I

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def random_walks(num_walks, num_steps):
5     walks = np.random.choice([-1, 1], size=(num_walks,
6     num_steps))
7     return walks
8
9 def scaled_positions(walks):
10     positions = np.sum(walks, axis=1)
11     return positions / np.sqrt(walks.shape[1])
12
13 num_walks = 10000
14 num_steps = 500
15
16 walks = random_walks(num_walks, num_steps)
17 scaled_pos = scaled_positions(walks)
```

Central Limit Theorem for One-dimensional Random Walks II

```
18 plt.hist(scaled_pos, bins=50, density=True, alpha=0.6, color  
    = 'g')  
19  
20 # Plot the standard normal distribution  
21 x = np.linspace(-4, 4, 100)  
22 y = (1 / np.sqrt(2 * np.pi)) * np.exp(-x**2 / 2)  
23 plt.plot(x, y, 'r-', linewidth=2)  
24  
25 plt.xlabel('Scaled Position')  
26 plt.ylabel('Frequency')  
27 plt.legend(['Normal Distribution', 'Random Walk'])  
28 plt.savefig('Example 1.pdf')
```

code/Example 1.py

Central Limit Theorem for One-dimensional Random Walks III

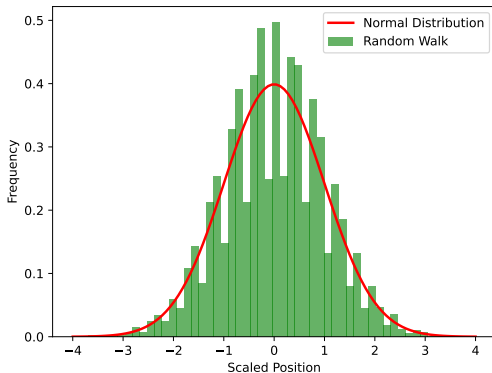


Figure: Central Limit Theorem for One-dimensional Random Walks.

Higher-dimensional Random Walks

Definition.

A random walker on the d -dimensional integer lattice

$$\mathbb{Z}^d = \{(x_1, x_2, \dots, x_d) \mid x_1, \dots, x_n \in \mathbb{Z}\}.$$

The random walker chooses one of its $2d$ closest neighbours, each with probability $\frac{1}{2d}$, and moves to that point at each time step. Let

$$S_n = x + X_1 + \dots + X_n$$

denote the position of the particle where $x, X_1, \dots, X_n, S_n$ represent points in $\mathbb{Z}^d$, i.e., they are d -dimensional vectors with integer components. The increments $X_1, X_2, \dots$ are unit vectors with one component of absolute value 1.

■ $\mathbb{E}[S_n] = 0.$



$$\mathbb{E}[|S_n|^2] = \mathbb{E}\left[\left(\sum_{j=1}^n X_j\right)\left(\sum_{k=1}^n X_k\right)\right] = n.$$

In d dimensions, the number of lattice points within distance $\sqrt{n}$ grows like $(\sqrt{n})^d$. As a result, the probability that we choose a particular point should decay like a constant times $n^{-\frac{d}{2}}$.



$$\mathbb{P}(S_{2n} = 0) \sim \frac{C_d}{n^{\frac{d}{2}}},$$

where

$$C_d = \frac{d^{\frac{d}{2}}}{\pi^{\frac{d}{2}} 2^{d-1}}.$$

Recurrent and Transient

Definition.

We say a random walk is **recurrent** if $\mathbb{P}(S_n = 0) = 1$ and **transient** if $\mathbb{P}(S_n = 0) < 1$.

Theorem.

Suppose S_n is a simple random walk in $\mathbb{Z}^d$ with $S_0 = 0$. If $d = 1, 2$, the random walk is recurrent, meaning that with probability one, it returns to the origin infinitely often. If $d \geq 3$, the random walk is transient, signifying that with probability one, it returns to the origin only finitely often. Furthermore,

$$\mathbb{P}(S_n \neq 0) > 0$$

$\forall n > 0$ if $d > 3$.

Proof. (One-dimension)

Definition.

Let us define $J_n = 1_{S_n=0}$. Let the **total number of visits** to the origin by the random walker be

$$V = \sum_{n=0}^{\infty} J_{2n}.$$

Since

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{E}[J_{2n}] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0).$$

Recalling

$$\mathbb{P}(S_{2n} = 0) \sim \frac{1}{\sqrt{\pi}} n^{-\frac{1}{2}}$$

as $n \rightarrow \infty$, we find that

$$\mathbb{E}[V] = \infty.$$

Example (St. Petersburg Paradox).

It is possible for a random variable to be finite yet have an infinite expectation. A well-known example of such a random variable arises from the St. Petersburg Paradox. Imagine playing a game where you flip a coin until you get tails. If you obtain k heads before flipping tails, your payoff is 2^k . The probability of getting exactly k heads is the probability of experiencing k consecutive heads followed by tails, which is $2^{-(k+1)}$. Consequently, the expected payoff in this game is

$$2^0 \cdot \frac{1}{2} + 2^1 \cdot \frac{1}{2^2} + 2^2 \cdot \frac{1}{2^3} + \dots = \frac{1}{2} + \frac{1}{2} + \dots + \frac{1}{2} = \infty.$$

Since the expectation is infinite, one should theoretically be willing to spend any amount of money to play this game once. However, this is not the case in reality, and herein lies the paradox.

Proof. (One-dimension)

Let q represent the probability that the random walker returns to the origin after time 0. Suppose $q < 1$, then $\mathbb{P}(V = 1) = 1 - q$, as $V = 1$ if and only if the walker never returns after time zero. More generally, we have

$$\mathbb{P}(V = k) = q^{k-1}(1 - q),$$

where $k \in \mathbb{N}^*$. This relationship implies that

$$\mathbb{E}[V] = \sum_{k=1}^{\infty} k\mathbb{P}(V = k) = \sum_{k=1}^{\infty} kq^{k-1}(1 - q) = \frac{1}{1 - q} < \infty;$$

however, we have previously established that $\mathbb{E}[V] = \infty$. Thus, it must be the case that $q = 1$. Therefore, the probability that a one-dimensional simple random walker returns to the origin infinitely often is one, which implies recurrence.

Proof. (Higher-dimensions)

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0).$$

We can express the expected number of visits as

$$\mathbb{E}[V] = \frac{1}{1-q},$$

where $q = q_d$ denotes the probability that the d -dimensional walk returns to the origin. Recalling that

$$\mathbb{P}(S_{2n} = 0) \sim \frac{C_d}{n^{\frac{d}{2}}},$$

we can now write the expected number of visits as

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0) = \begin{cases} < \infty & \text{if } d \geq 3 \\ = \infty & \text{if } d = 2 \end{cases}.$$



One-dimensional Boundary Value Problems

Definition.

Suppose N is a positive integer and a random walker starts at $x \in \{0, 1, \dots, N\}$. Let S_n denote the position of the walker at time n . The walker stops when it reaches either 0 or N , and we define the **stopping time** as

$$T = \min\{n \mid S_n = 0 \text{ or } N\}.$$

Then, we define the function $F : \{0, 1, \dots, N\} \rightarrow [0, 1]$ by

$$F(x) = \mathbb{P}(S_T = N \mid S_0 = x)$$

which represents the probability that a random walker starts at position x and ultimately reaches the boundary point N .

The stopping time should be finite since the function $F(x)$ would be rendered undefined if S_T were left undetermined.

Example (Gambler's Ruin).

The player begins with x chips and persists in playing until they either amass N chips or exhaust their funds. The probability that the gambler evades bankruptcy before attaining N is given by $F(x)$. Evidently, $F(0) = 0$ and $F(N) = 1$. Suppose $0 < x < N$. After the first game, the gambler has either $x - 1$ or $x + 1$ chips, and both of these outcomes bear equal likelihood. Consequently,

$$F(x) = \frac{1}{2}F(x+1) + \frac{1}{2}F(x-1), \quad (4)$$

where $x = 1, \dots, N-1$. One function F that satisfies equation (4) with the **boundary conditions** $F(0) = 0$, $F(N) = 1$ is the linear function

$$F(x) = \frac{x}{N}.$$

Theorem.

Suppose a, b are real numbers and N is a positive integer. Then the only function $F : \{0, \dots, N\} \rightarrow \mathbb{R}$ satisfying equation (4) with the boundary conditions $F(0) = a$ and $F(N) = b$ is the linear function

$$\tilde{F}(x) = a + \frac{x(b-a)}{N}.$$

Uniqueness

- Consider a linear function $\tilde{F}$ satisfying the boundary conditions, with $\tilde{F}(0) = a$ and $\tilde{F}(N) = b$. Suppose F is another solution to the boundary value problem described by equation (4) that is distinct from $\tilde{F}$. For each $0 < x < N$, we have

$$F(x) \leq \max\{F(x-1), F(x+1)\}.$$

- This inequality implies that $F(x)$ cannot possess local maxima within the interior of the domain $(0, N)$. Consequently, the only possible locations for the maximum of the function F must be at the boundary points $x = 0$ and $x = N$. By analogous reasoning, we deduce that the minimum of F is also achieved on the boundary $\{0, N\}$.
- Suppose $F(0) = a$ and $F(N) = b$. We define a new function $F' = F - \tilde{F}$. By construction, F' satisfies equation (4) with $F'(0) = 0$ and $F'(N) = 0$. From our analysis in the previous case, F' must be identically zero. Thus, we obtain the desired result $F = \tilde{F}$, which is the unique solution. □

Higher-dimensional Boundary Value Problems

Definition.

$$\partial A = \{z \in \mathbb{Z}^d \setminus A \mid \text{dist}(z, A) = 1\}.$$

With this in place, we let $\bar{A} = A \cup \partial A$ denote the **closure** of A , encapsulating both the interior and the boundary. In this context, the set A is also referred to as the **interior** of $\bar{A}$.

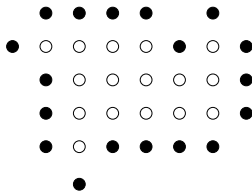


Figure: The white dots are A and the black dots are ∂A .

Three-dimensional Random Walks with Boundaries I

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from mpl_toolkits.mplot3d import Axes3D
4
5 def random_walk_3d(steps, boundary):
6     # Initialize the starting position at (0, 0, 0)
7     x, y, z = 0, 0, 0
8     walk_x, walk_y, walk_z = [x], [y], [z]
9
10    # Iterate through the number of steps
11    for _ in range(steps):
12        # Choose a random axis (0, 1, or 2) and step
13        # direction (-1 or 1)
14        axis = np.random.choice([0, 1, 2])
15        step = np.random.choice([-1, 1])
16
17        # Update the position based on the chosen axis
18        if axis == 0:
19            x_new, y_new, z_new = x + step, y, z
20        elif axis == 1:
21            x_new, y_new, z_new = x, y + step, z
```

Three-dimensional Random Walks with Boundaries II

```
21     else:
22         x_new, y_new, z_new = x, y, z + step
23
24     # Check if the new position is within the boundaries
25     if (-boundary <= x_new <= boundary and
26         -boundary <= y_new <= boundary and
27         -boundary <= z_new <= boundary):
28
29         # Update the current position
30         x, y, z = x_new, y_new, z_new
31
32         # Add the new position to the walk
33         walk_x.append(x)
34         walk_y.append(y)
35         walk_z.append(z)
36
37     return walk_x, walk_y, walk_z
38
39 steps = 1000
40 boundary = 10
41
42 walk_x, walk_y, walk_z = random_walk_3d(steps, boundary)
```

Three-dimensional Random Walks with Boundaries III

```
43  
44 # Create a 3D plot  
45 fig = plt.figure(figsize=(10, 10))  
46 ax = fig.add_subplot(111, projection='3d')  
47  
48 # Plot the 3D random walk  
49 ax.plot(walk_x, walk_y, walk_z, marker='o', markersize=3,  
         linestyle='-')  
50 ax.set_xlim(-boundary-1, boundary+1)  
51 ax.set_ylim(-boundary-1, boundary+1)  
52 ax.set_zlim(-boundary-1, boundary+1)  
53 ax.set_xlabel('X')  
54 ax.set_ylabel('Y')  
55 ax.set_zlabel('Z')  
56 plt.grid(True)  
57 plt.savefig('Example 3.pdf')
```

code/Example 3.py

Three-dimensional Random Walks with Boundaries IV

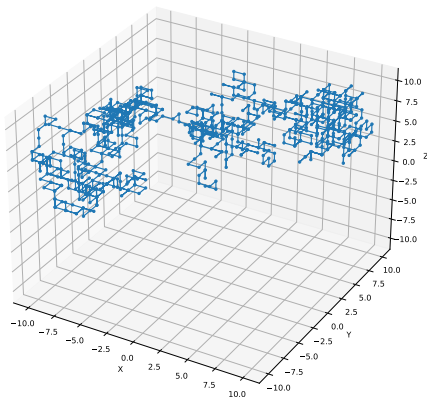


Figure: Three-dimensional Random Walks with Boundaries.

Definition.

Let A be a subset of $\mathbb{Z}^d$. A function $F : \bar{A} \rightarrow \mathbb{R}$ is said to possess the **mean value property** if for all $x \in A$,

$$F(x) = \frac{1}{2d} \sum_{\substack{y \in \mathbb{Z}^d \\ |x-y|=1}} F(y).$$

Define a **linear operator** $\mathcal{L}$ acting upon a function $F : \mathbb{Z}^d \rightarrow \mathbb{R}$ by

$$\mathcal{L}F(x) = \frac{1}{2d} \sum_{\substack{y \in \mathbb{Z}^d \\ |x-y|=1}} [F(y) - F(x)].$$

This operator is known as the **(discrete) Laplacian**. A function F is deemed **(discrete) harmonic** at a point x if $\mathcal{L}F(x) = 0$. This boundary value problem also referred to as the **Dirichlet problem** for harmonic functions.

Dirichlet Problem for Harmonic Functions

Definition.

Given a set $A \subset \mathbb{Z}^d$ and a function $F : \partial A \rightarrow \mathbb{R}$, we seek an extension $\tilde{F} : \bar{A} \rightarrow \mathbb{R}$ of F such that F is harmonic in A . In other words, we want to satisfy the Laplace equation

$$\mathcal{L}F(x) = 0 \tag{5}$$

for all $x \in A$.

Definition (Stopping Time).

Consider a random walk S_n in $A \subset \mathbb{Z}^d$. We define the **stopping time** as

$$T_A = \min\{n \geq 0 \mid S_n \notin A\},$$

representing the moment when the random walker first reaches the boundary of the set A .

Theorem.

If $A \subset \mathbb{Z}^d$ is finite, then for every $F : \partial A \rightarrow \mathbb{R}$, there is a unique extension of F to $\bar{A}$ such that it satisfies equation (5). It is given by

$$\tilde{F}(x) = \mathbb{E}[F(S_{T_A}) \mid S_0 = x] = \sum_{y \in \partial A} \mathbb{P}(S_{T_A} = y \mid S_0 = x) F(y).$$

Proof. We can consider a spatial dimension to verify that $\tilde{F}$ is indeed a solution to the Dirichlet problem. The real challenge lies in demonstrating the uniqueness of this solution.

Uniqueness

- Suppose F is harmonic on A , $S_0 = x \in \bar{A}$, and let

$$M_n = F(S_{n \wedge T_A}),$$

where $n \wedge T_A$ means the minimum of n and T_A . Then equation (5) can be rewritten as

$$\mathbb{E}[M_{n+1} \mid S_0, \dots, S_n] = F(S_{n \wedge T_A}) = M_n.$$

A process that satisfies $\mathbb{E}[M_{n+1} \mid S_0, \dots, S_n] = M_n$ is called a **martingale (with respect to the random walk)**. $F(S_{n \wedge T_A})$ being a martingale is essentially equivalent to F being harmonic on A .

- The martingales satisfy $\mathbb{E}[M_n] = \mathbb{E}[M_0]$, and hence, if $S_0 = x$, then

$$\sum_{y \in \bar{A}} \mathbb{P}(S_{n \wedge T_A} = y) F(y) = \mathbb{E}[M_n] = M_0 = F(x).$$

Taking limits, we obtain

$$F(x) = \lim_{n \rightarrow \infty} \sum_{y \in \bar{A}} \mathbb{P}(S_{n \wedge T_A} = y) F(y) = \sum_{y \in \partial A} \mathbb{P}(S_{T_A} = y) F(y).$$



- However, if we allow A to be an infinite set, the uniqueness of the solution is no longer guaranteed. Consider a one-dimensional example with $A = \{1, 2, 3, \dots\}$ and $\partial A = \{0\}$. In this case, for every $c \in \mathbb{R}$, the function $F(x) = cx$ is harmonic in A with a boundary value of 0 at the origin.
- Suppose F is harmonic on A with $F(0) = 0$, and let S_n be a simple random walk starting at a positive integer x . As before, we let $T = \min\{n \geq 0 \mid S_n = 0\}$ and $M_n = F(S_{n \wedge T})$. The same argument shows that M_n is a martingale and

$$F(x) = \mathbb{E}[M_n] = \sum_{y=0}^{\infty} F(y) \mathbb{P}(S_{n \wedge T} = y).$$

- With probability one, $T < \infty$. Then $\mathbb{P}(S_{n \wedge T} = 0)$ tends to 1, which can be expressed as

$$\lim_{n \rightarrow \infty} \sum_{y>0} \mathbb{P}(S_{n \wedge T} = y) = 0.$$

- However, if F is unbounded, we cannot conclude from this that

$$\lim_{n \rightarrow \infty} \sum_{y > 0} F(y) \mathbb{P}(S_{n \wedge T} = y) = 0.$$

Nonetheless, our exploration reveals that there is only one bounded function that is harmonic on A with a given boundary value at 0.

Theorem.

Suppose A is a proper subset of $\mathbb{Z}^d$ such that for all $x \in \mathbb{Z}^d$,

$$\lim_{x \rightarrow \infty} \mathbb{P}(T_A > n \mid S_0 = x) = 0.$$

Suppose $F : \partial A \rightarrow \mathbb{R}$ is a bounded function. Then there is a unique bounded extension of F to $\bar{A}$ that satisfies equation (5). It is given by

$$\tilde{F}(x) = \mathbb{E}[F(S_{T_A}) \mid S_0 = x] = \sum_{y \in \partial A} \mathbb{P}(S_{T_A} = y \mid S_0 = x) F(y).$$

Conclusion

- We began our exploration with a one-dimensional random walk, wherein we rigorously demonstrated that its probability distribution adheres to the **Central Limit Theorem**.
- Random walkers exhibit **recurrent** tendencies in one- and two-dimensional scenarios, but **transient** characteristics emerge in higher-dimensional cases.
- In the one-dimensional case, we discovered a **unique solution** to **boundary value problems**.
- **Main conclusion**: it is indeed possible to find a unique function that extends an existing function—defined only on the boundary of a specific region, whether finite or not—to the entire region. In doing so, we effectively unravelled the **Dirichlet problem**.

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Thank You !